MILITARY SPECIFICATION

DRIVER, EXPLOSIVE, MARK 3 MOD 0

This specification has been approved by the
Naval Ordnance Systems Command, Department of the Navy.

1. Scope. This specification establishes the requirements for the procurement of the Driver, Explosive, Mark 3 Mod 0, Naval Ordnance Systems Command LD 479733. This driver is electrically initiated. The Driver, Explosive, Mark 3 Mod 0 is of one type and class.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings and publications and such portions thereof as designated herein of the issue in effect on the date of invitation for bids form a part of this specification:

SPECIFICATIONS

FEDERAL

RR-S-366	Sieve, Standard for Testing Purposes
TT-B-838	Butyl Acetate, Normal (For Use in Organic Coatings)
PPP-B-566	Boxes, Folding, Paperboard
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner
PPP-B-676	Boxes, Set-up, Paperboard
PPP-B-1055	Barrier Material, Waterproofed, Flexible

MILITARY

MIL-A-148	Waterproof, Flexible Aluminum Foil

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STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-286	Propellants, Solid; Sampling, Examination and Testing
MIL-STD-331	Fuze and Fuze Components, Environmental and Performance Tests for
MIL-STD-454	Standard General Requirements For Electronic Equipment
MIL-STD-481	Configuration Control - Engineering Changes, Deviations and Waivers (Short Form)

DRAWINGS

NAVAL ORDNANCE SYSTEMS COMMAND

LD 479733	Driver, Explosive, Mark 3 Mod 0
LD 480002	Jolt & Vibration Test Fixture
LD 480047	Drop Test Fixture
LD 486146	Output Test Fixture

And all drawings, standards and specifications listed thereon. In the following specification, all referenced drawing numbers are listed on LD 479733.

PUBLICATIONS

NAVAL ORDNANCE SYSTEMS COMMAND

OD 6652	Preparation of Diazodinitrophenol Ignition Mixture for Use in Electric Primers
OD 6699	Advisory Procedure for Processing Normal Lead Styphnate for Use in Sensitive Ignition Charges
WR-43	Naval Weapons Requirements, Preparation of Quality Assurance Provisions

FEDERAL

Federal Test Method No. 406	Plastics, Method of Testing

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(Copies of specifications, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Description. The Driver, Explosive, Mark 3 Mod 0 is an element used to provide external mechanical action and is initiated electrically.

3.2 General requirements.

3.2.1 First article sample. Unless otherwise specified in the contract or order, a first article sample of the explosive drivers shall be manufactured using the methods and procedures proposed for the production lot. The sample shall be prepared for delivery in accordance with Section 5 herein and will be tested as specified in Section 4 herein. The sample is for the purpose of determining that, prior to starting production, the contractor's production methods are capable of producing items that comply with this specification and the design. Provision for the sample shall be in accordance with 4.3.1.

3.2.2 Materials. Unless otherwise specified in the contract, requisition or order, all materials and purchased parts entering into the final assembly of the Driver, Explosive, Mark 3 Mod 0 shall conform strictly to the detailed requirements indicated on the drawings and specifications listed in LD 479733.

3.2.3 Finishes, coatings, construction and dimensions. Finishes, coatings, construction and dimensions for the Driver, Explosive, Mark 3 Mod 0 shall be strictly in accordance with the applicable drawings, specifications and publications listed in LD 479733.

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3.2.4 Marking. Marking of assemblies, subassemblies and parts shall be in accordance with the applicable drawings listed in LD 479733.

3.2.5 Measuring devices. The contractor shall provide himself with necessary and adequate measuring devices to insure that the material to which this specification applies will meet the dimensional requirements shown on the applicable drawings.

3.2.6 Requests for deviation. Requests for deviation from this specification, applicable drawings, specifications, publications or drawings listed herein shall be in accordance with MIL-STD-481.

3.2.7 Conflicting requirements. Conflicting requirements arising between this specification and any specifications, publications or drawings listed herein shall be referred in writing to the procuring agency or appointed agent for interpretation, clarification and resolution.

3.2.8 Data. No data are required by applicable documents referenced in Section 2 unless specified in the contract or order.

3.3 Performance requirements and product characteristics. The Driver, Explosive, Mark 3 Mod 0 shall meet the following performance requirements and product characteristics:

3.3.1 Plug and lead wire assembly.

3.3.1.1 Acetone extraction. The resinous component of the assembly shall be thoroughly cured and shall have an acetone extract value of 5.0 percent maximum. The cured plug shall have a smooth, glossy surface free from checks or cracks. Inspection and testing procedures shall be in accordance with 4.5.2.1.

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3.3.1.2 Insulation resistance. The insulation resistance between the twisted lead wires and the cup shall be a minimum of 50 megohms when tested in accordance with 4.5.2.6.

3.3.1.3 Lead wire securement. The lead wires shall be free from kinks and nicks and shall be capable of withstanding a pull along the axis of the plug sufficient to break the leads without showing any signs of slipping in the plug when tested in accordance with 4.5.2.3.

3.3.1.4 Lead wire flexibility. Each lead wire shall be capable of withstanding, without breaking, successive bends in opposite directions of 90°, 180° and 90° when tested in accordance with 4.5.2.4.

3.3.2 Bridged plug assembly.

3.3.2.1 Soldering. Solder in accordance with MIL-STD-454, Requirement 5.

3.3.3 Ignition assembly.

3.3.3.1 Sleeve. When assembling the sleeve, care shall be exercised to assure that neither the bridge material nor other parts of the subassembly are injured.

3.3.3.2 Flash charge. The flash charge mixture shall conform to the requirements of Drawing 1706087 which is part of LD 479733. The testing of the nitrostarch used in the flash charge shall be in accordance with 4.5.2.2. See OD 6652 for instruction on the preparation of this mixture.

3.3.3.3 Charge composition. The charge composition shall conform to the requirements of Drawing 1825768 which is part of LD 479733. The testing of the nitrostarch used in the charge composition shall be in accordance with 4.5.2.2.

3.3.4 Driver, Explosive, Mark 3 Mod 0 Assembly.

3.3.4.1 Circuit resistance. The resistance of the finished primer before shunting the leads shall be between 3 and 7 ohms. Testing and acceptance shall be in accordance with 4.5.2.5.

3.3.4.2 Insulation. The insulation resistance between the shunted lead wires and the driver body shall not be less than 50 megohms. Testing and acceptance shall be in accordance with 4.5.2.6.

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3.3.4.3 Functioning. The explosive driver shall fire within 0.030 second maximum when a 1.0 ± 0.2 percent microfarad capacitor charged to 38.5 ± 0.2 volts is discharged through the bridge wire circuit. Test in accordance with 4.5.2.7.

3.3.4.4 Output. When fired as specified in 4.5.2.8, the driver shall provide a minimum motion of 0.110 inch against a load of 8 lbs. minimum without rupture of the cup. Testing and acceptance shall be in accordance with 4.5.2.8.

3.3.4.5 Moisture sealing. The driver shall meet the requirements of 3.3.4.3 and 3.3.4.4 after 6 hours water immersion under a pressure of 25 psi gage at 75 ± 5°F. Testing and acceptance shall be in accordance with 4.5.2.9.

3.3.5 Moisture content.

3.3.5.1 Ignition and case assemblies. The ignition and case assemblies shall be stored in desiccators or dry houses until ready for final assembly.

3.3.6 Loading.

3.3.6.1 Ignition assembly loading.

3.3.6.1.1 Flash charge application. The flash charge shall be applied over the bridge wire by means of a suitable nonsparking device of such characteristics as to allow approximately 0.001 gram of the ignition charge to remain on the bridge surface. To apply the charge over the bridge, the applicator shall be placed in the mixture in the dish, and the proper amount of material picked up. The material is transferred to the assembly by touching the drop of mixture to the bridge. The ignition charge shall be dried to an extent which provides sufficient mechanical strength to withstand the displacement forces incident to loading the charge. A minimum drying time of 4 hours at 120°F is recommended. The flash charge shall be processed and handled in such a manner as to preclude the presence of moisture immediately prior to the addition of the charge. For advisory instruction on handling ignition charges see 6.4.

3.3.6.1.2 Charge loading. The technique of loading the charge shall be such as to prevent displacement of the flash charge or damage to other parts.

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3.3.7 Storage of ignition assembly. The ignition assembly shall be stored in desiccators or dry houses until ready for final assembly.

3.4 Durability requirements.

3.4.1 Vibration. The driver shall be mounted on a vibration device and shall meet the following requirements:

3.4.1.1 Vibration, ready nonoperating. The driver, less packaging, shall be mounted and subjected to the test specified in 4.6.1.1 and thereafter shall show no damage and shall meet the requirements of 3.2.3, 3.3.4.1, 3.3.4.2, 3.3.4.3, 3.3.4.4 and 3.3.4.5 of this specification.

3.4.1.2 Transportation and handling. The drivers packaged and packed as defined in Section 5 shall be subjected to tests as specified in 4.6.1.2 and thereafter shall show no damage and shall meet the requirements of 3.2.3, 3.3.4.1, 3.3.4.2, 3.3.4.3, 3.3.4.4 and 3.3.4.5 of this specification.

3.4.2 Shock, free fall. The drivers packaged and packed as defined in Section 5 shall be subjected to tests as specified in 4.6.2 and thereafter shall show no damage and shall meet the requirements of 3.2.3, 3.3.4.1, 3.3.4.2, 3.3.4.3, 3.3.4.4 and 3.3.4.5 of this specification.

3.4.3 Temperature cycle. The loaded explosive drivers shall meet acceptably the requirements of 3.3.4.3 and 3.3.4.4 after being subjected to the temperature cycle test of 4.6.3.

3.4.4 Extreme temperature functioning. The explosive driver shall be capable of firing within 0.030 second maximum at 28° ± 2°F or 90° ± 3° F when a 1.0 ± 0.2 percent microfarad capacitor charged to 38.5 ± 0.1 volts is discharged through the bridge wire circuit. Test in accordance with 4.6.4.

3.4.5 Safety requirements.

3.4.5.1 Jolt. The driver shall be mounted and subjected to jolt tests as specified in 4.6.5 and thereafter shall not have fired and shall be safe to handle and dispose of.

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3.4.5.2 Forty-foot drop. The driver shall be mounted and subjected to the forty-foot drop test as specified in 4.6.6 and thereafter shall not have fired and shall be safe to handle and dispose of.

3.5 Cleanup. Prior to and after final assembly, all parts, components and the assembly shall be thoroughly cleaned of loose metal chips and other foreign matter. Burrs and sharp edges shall be removed.

3.6 Workmanship. The equipment, including all parts and accessories, shall be constructed and finished in a manner to assure compliance with all requirements of this specification. Particular attention shall be paid to neatness and thoroughness of soldering, plating, crimping, marking of parts and assemblies and freedom of parts from burrs and sharp edges. The standard of workmanship exhibited in any approved first article sample, subject to any qualification stated in the Government's notice of approval, shall determine the requirements of the contract relative to workmanship insofar as not specifically covered by applicable specifications.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. Inspection of the explosive drivers shall be classified as follows:

a. First article inspection (Paragraphs 4.5 and 4.6)

b. Quality conformance inspection (Paragraph 4.5)

c. Periodic sampling inspection (Paragraphs 4.5 and 4.6)

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4.3 Inspection.

4.3.1 First article sample. A first article sample of explosive drivers manufactured in accordance with 3.2.1 shall be subjected to the examinations and tests detailed in 4.5 and 4.6 at an activity designated by the procuring agency. Acceptance of the sample shall be based on no defects in the sample. Further production of the explosive drivers by the supplier, prior to the approval of the sample, shall be at the supplier's risk. See TABLE I.

4.3.2 Quality conformance.

4.3.2.1 100 percent inspection. Each unit shall be subjected to the examinations of 4.5.1 and tests designated as 100 percent in 4.5.2. Any unit containing one or more defects shall be cause for regarding the unit as defective.

4.3.2.2 Sampling. Tests designated as sample tests in 4.5.2 shall be conducted on a random sample of explosive drivers selected from each production lot in accordance with the provisions of MIL-STD-105, inspection level and AQL as indicated. No lot shall be shipped until the samples representing the lot have successfully passed the quality conformance tests.

4.3.2.2.1 Classification of characteristics. The characteristics verified by the tests and examinations herein are classified as Critical, Major or Minor in accordance with WR-43. Tests and examinations that verify critical characteristics are identified by the symbol (C), and major characteristics, by the symbol (M). The number following the classification symbol indicates the serial number of the test or examination. Tests and examinations which are not annotated with a classification code are classified minor.

4.3.3 Periodic sampling. From each production lot of explosive drivers which have passed the acceptance tests, a sample of explosive drivers will be selected at random for submission to an activity designated by the procuring agency to determine compliance with the requirements of the contract, specifications and drawings. The periodic production sample shall be considered satisfactory if the explosive drivers meet all the specification and drawing requirements and pass all the tests of 4.5 and 4.6. See TABLE III.

4.3.4 Test conditions. Unless otherwise specified, the test procedure of 4.5.2 shall be accomplished under the following conditions:

4.3.4.1 Temperature: Room ambient 60°F to 95°F.

4.3.4.2 Humidity: Room ambient to 95 percent relative maximum.

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4.4 Special test equipment. The following items of test equipment are required to perform the quality conformance tests specified by 4.5.2:

a. A resistance measuring device capable of measuring at least 50 megohms at ± 5 percent accuracy with a 500-volt ± 50 volts dc test potential.

b. An ohmmeter or resistance bridge capable of measuring 3.0 ohms to 7.0 ohms with an accuracy of 0.1 ohm. Current must be limited to 5 milliamperes maximum.

c. A firing circuit as specified by Figure 3.

d. A 214 gram weight.

e. A set of U. S. Standard sieves complying with Federal Specification RR-S-366 consisting of a cover, a No. 40, No. 80 and No. 140 sieves and a receiving pan.

f. A balance capable of weighing accurately to ± 0.5 milligram.

g. A nitrometer as shown by MIL-STD-286, Method 209.3.2 Figure 1.

h. An acetone extraction apparatus as specified by Federal Test Method Std. No. 406, Method 7021.

i. A heating unit suitable for heating the acetone extraction apparatus.

j. A drying oven in which temperatures of 122°F ± 3.6°F and 212°F ± 10°F can be maintained with sufficient draft to prevent explosion of acetone fumes.

k. A frequency counter capable of measurements to 0.001 second, accuracy ± 0.0001 second.

4.5 Quality conformance inspection.

4.5.1 Examination. Each explosive driver shall be examined in accordance with 4.5.1.1 and 4.5.1.2.

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4.5.1.1 Soldering examination. Electrical connections shall be examined prior to soldering to determine that all connections are mechanically secure. After soldering, the unit shall be examined to determine that there are no cold solder joints and that all excess flux and all solder spatter have been removed.

4.5.1.2 Cleanliness examination. Each explosive driver shall be examined for cleanliness. The presence of any visible moisture, corrosion or other contamination, of metal chips, loose, excess or spattered solder, resin flash, burrs or sharp edges shall cause the explosive actuator to be regarded defective.

4.5.2 Tests.

CAUTION

TO AVOID POSSIBLE INJURY TO PERSONNEL,
EXPLOSIVE DRIVERS SHALL BE CONTAINED IN
A SUITABLE SAFETY CONTAINER DURING TESTS

4.5.2.1 Acetone extraction test of molded plug and lead wire assembly (M101). At least one test of each lot and additional tests, where necessary, to yield one sample for each batch of uncured resin or major change in curing procedure shall be made. Select a sufficient sample of molded plugs to conduct the acetone extraction test as prescribed in Method No. 7021, Federal Test Method Std. No. 406. In cases where the extract exceeds 5 percent, all plugs represented by the sample shall be rejected.

4.5.2.2 Chemical properties test of nitrostarch.

4.5.2.2.1 Nitrogen content test (M102). The nitrometer method shall be used to determine the nitrogen content of the nitrostarch; sampling and chemical analysis shall be in accordance with MIL-STD-286, Method 209.3.2. The nitrostarch shall meet the nitrogen requirements of Drawing 1825768, a part of LD 479733.

4.5.2.2.2 Stability test (M103). A representative sample of air-dried nitrostarch shall be dried at 100° C for 20 minutes. Introduce a 0.6 gram sample of the dried nitrostarch into a standard test tube approximately 75 mm long and 10 mm in diameter. Heat the test tube and contents at 100° C ± 0.5° C for 48 hours in an ambient atmosphere having a maximum absolute humidity of .008 gram of water per gram of dry air (60 percent relative humidity at 65° F). When a sample of nitrostarch ignites or explodes during the stability test, the batch of nitrostarch from which the sample was taken shall not be used in the preparation of the flash charge.

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4.5.2.3 Lead wire securement test (Sampling) (M104). Plugs shall be selected and tested from each lot in accordance with MIL-STD-105, Inspection Level S4, AQL 1.0. The lead wires of an acceptable plug shall withstand a pull along the axis of the plug sufficient to break the leads without showing any signs of slipping in the plug. The rate of pull or loading shall not exceed 0.5 inch per minute. Disposition of the lot shall be in accordance with the above sampling plan.

4.5.2.4 Lead wire flexibility test (Sampling) (M105). Drivers shall be selected and tested from each lot in accordance with MIL-STD-105, Inspection Level S4, AQL 1.0 percent defective. Each lead shall withstand, without breaking, successive bends in opposite directions of 90°, 180° and 90° while a tensile force of 214 grams is applied to the wire. The bending shall occur as close as possible to the plug face in a plane normal to the plug face using the point of exit as a fulcrum and shall be at the rate of approximately 3 seconds per bend.

4.5.2.5 Driver circuit electrical resistance test (C1) (100%). Before shunting the lead wires, the resistance of the bridge wire shall be tested with the ohmmeter, Item 4.4b. The testing current in the bridge circuit of the driver shall be less than 5 milliamperes during tests. Drivers having a resistance outside the limits of 3 and 7 ohms shall be classed as defective.

4.5.2.6 Insulation resistance test of driver assembly (C2)(100%). Using the shunted lead wires and driver cup as terminals, measure the insulation resistance between them with the Resistance Measuring Device, Item 4.4a. To expedite production, insulation resistance may be measured at any temperature above 70°F and any relative humidity above 70 percent. An explosive driver successfully meeting the requirements of 3.3.4.2 when tested in accordance with the above conditions may be accepted. When an explosive driver fails to meet requirements under these conditions, it may be conditioned at a temperature of 77°F ± 2°F and a relative humidity of 75 percent ± 5 percent for at least 24 hours. Within one-half hour after conditioning, explosive drivers shall be tested as specified above at any relative humidity above 60 percent. At the option of the activity (supplier or Government) performing the tests, all explosive fittings may be conditioned before testing.

4.5.2.7 Driver functioning (Sampling)(M106). The test circuit of Figure 3 shall be used to deliver the firing pulse. The testing temperature shall be 75 ± 5°F. Delay time shall be ascertained to 0.001 second when measured with the Frequency Counter, Item 4.4k. The circuit for initiating pulses should have a high impedance input which dissipates less than 1.0 percent of the test circuit output. The leads of the firing circuit should be as short as convenience permits. Sampling plans are as follows:

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a. Sampling plans for functioning without consideration of delay time feature:

Sample size in accordance with MIL-STD-105, Inspection Level I, except minimum sample size - 20 units.

Sample Size	Acceptance No.	Rejection No.
200 units or less	0	1
315 or 500 units	1	2
800 units	2	3
1,250 units	3	4
2,000	5	6

b. Sampling plans for delay time feature:

Sample size in accordance with MIL-STD-105, Special Inspection Level S-4, except minimum sample size = 13 units.

Sample Size	Acceptance No.	Rejection No.
13 or 20 units	0	1
32 or 50 units	1	2
80 units	2	3
125 units	3	4
200 units	5	6
315 units	7	8
500 units	10	11
800 units	14	15
1,250 or 2,000 units	21	22

4.5.2.8 Output test (Sampling) (M107). Drivers shall be selected and tested from each lot in accordance with MIL-STD-105, Inspection Level S4, and an AQL of 1.0 percent defective. The drivers are required to move an eight-pound load a minimum of 0.110 inch without rupture of the cup when initiated in the output test fixture LD 486146. Drivers which fail to meet these requirements shall be classed as defective. Disposition of the lot shall be in accordance with the sampling plan given above.

4.5.2.9 Moisture sealing test (Sampling) (M108). The explosive drivers shall be subjected to 6 hours of water immersion under a pressure of 25 psi gage at a temperature of 75° ± 5°F. After completion of the water immersion period, removal from the water and thorough drying of the exterior insulation, the drivers shall be capable of meeting the requirements of 3.3.4.3 and 3.3.4.4. Sample in accordance with MIL-STD-105, Inspection Level, S4, AQL 1.0 percent defective.

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TABLE I

Allocation of first article samples for testing

Test	Paragraph	Site of Test	No.of Units	Notes
Molded plug and lead wire	4.5.2.1 (Destructive)	Manuf. plant	As required herein	
Nitrostarch	4.5.2.2 (Destructive)	Manuf. plant	As required herein	
Lead wire (molded plug)	4.5.2.3 (Destructive)	Manuf. plant	10	
Lead wire flexibility	4.5.2.4 (Destructive)	Manuf. plant	20	
Bridge wire resistance	4.5.2.5 (Nondestructive)	Manuf. plant	20	This sample shall be subsequently used to fill sample requirements of 4.5.2.6
Insulation resistance	4.5.2.6 (Nondestructive)	Manuf. plant	20	This sample shall be subsequently used to fill sample requirements of 4.5.2.7, 4.5.2.8, 4.5.2.9 and 4.6.
Function	4.5.2.7 (Destructive)	Manuf. plant	20	
Output	4.5.2.8 (Destructive)	Manuf. plant	13	
Moisture sealing	4.5.2.9 (Nondestructive)	Manuf. plant	20	

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TABLE II

First article sample tests on the explosive driver to be performed at an activity designated by the procuring agency

Test	Paragraph	No. of Units
Examination Bridge wire resistance	4.5.1 4.5.2.5 (Nondestructive)	Designated by contract or procuring agency
Insulation resistance	4.5.2.6 (Nondestructive)	Designated by contract or procuring agency
Function	4.5.2.7 (Destructive)	Designated by contract or procuring agency
Output	4.5.2.8 (Destructive)	Designated by contract or procuring agency
Moisture sealing	4.5.2.9 (Nondestructive)	Designated by contract or procuring agency
Vibration, ready nonoperating	4.6.1.1 (Destructive)	Designated by contract or procuring agency
Vibration, transportation and handling	4.6.1.2 (Destructive)	Designated by contract or procuring agency
Free fall drop test	4.6.2 (Destructive)	Designated by contract or procuring agency
Temperature and humidity	4.6.3 (Destructive)	Designated by contract or procuring agency
Extreme temperature functioning	4.6.4 (Destructive)	Designated by contract or procuring agency
Jolt test	4.6.5 (Destructive)	Designated by contract or procuring agency
Forty foot drop test	4.6.6 (Destructive)	Designated by contract or procuring agency

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TABLE III

Periodic sample tests on the explosive driver to be performed at an activity designated by the procuring agency

Test	Paragraph	No. of Units
Examination	4.5.1	Designated by contract or procuring agency
Bridge wire resistance	4.5.2.5 (Nondestructive)	
Insulation resistance	4.5.2.6 (Nondestructive)	Designated by contract or procuring agency
Function	4.5.2.7 (Destructive)	Designated by contract or procuring agency
Output	4.5.2.8 (Destructive)	Designated by contract or procuring agency
Moisture sealing	4.5.2.9 (Nondestructive)	Designated by contract or procuring agency
Vibration, ready nonoperating	4.6.1.1 (Destructive)	Designated by contract or procuring agency
Vibration, transportation and handling	4.6.1.2 (Destructive)	Designated by contract or procuring agency
Free fall drop test	4.6.2 (Destructive)	Designated by contract or procuring agency
Temperature and humidity	4.6.3 (Destructive)	Designated by contract or procuring agency
Extreme temperature functioning	4.6.4 (Destructive)	Designated by contract or procuring agency
Jolt test	4.6.5 (Destructive)	Designated by contract or procuring agency
Forty-foot drop test	4.6.6 (Destructive)	Designated by contract or procuring agency

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4.6 Durability tests.

4.6.1 Vibration. The driver shall be subjected to vibration tests as specified below.

4.6.1.1 Vibration, ready nonoperating (M109). The explosive drivers shall be mounted on the Vibration Test Fixture, LD 480002, which shall be rigidly attached to the vibration table. The explosive drivers shall then be subjected to the vibration test per Test Procedure II, Method 104, MIL-STD-331.

4.6.1.2 Transportation and handling (M110). The explosive drivers packaged and packed as specified in Section 5 shall be placed on the test table and restrained only from falling off the table by suitable fences. The test table shall be oscillated so that it moves in a circular path in the vertical plane with a peak-to-peak amplitude of one inch. The frequency of the vibration shall be set at the point where the package of drivers jostles against the fence and momentarily leaves the table by about 3/16 inch. This frequency may be between four and five H_z. The pack shall be vibrated on each of the six faces for a period of 30 minutes. The test requires a total of three hours.

4.6.2 Free fall drop test (M111). The explosive drivers packaged and packed as indicated in Section 5 shall be subjected to a free fall drop shock from a height of 30 inches to a concrete or hard steel surface. This test shall be applied to each corner of the packed unit.

4.6.3 Temperature cycling (M112). Sample explosive drivers packaged in accordance with Section 5 shall be subjected to one 14-day JAN cycle as defined in MIL-STD-331, Test 105, except that the relative humidity need not be raised above room ambient or ambient as controlled. After being subjected to the temperature cycle, the sample explosive drivers shall successfully meet the requirements of 3.3.4.3 and 3.3.4.4.

4.6.4 Extreme temperature functioning (M113). The test circuit of Figure 3 shall be used to deliver the firing pulse. The delay time shall be within 0.030 second maximum when tested with the Frequency Counter, Item 4.4k. The circuit for initiating pulses should have a high impedance input which dissipates less than 1.0 percent of the test set output. The leads of the firing circuit should be as short as convenience permits. The explosive drivers shall be tested at a temperature of $28^{\circ} \pm 2^{\circ}$F or $90^{\circ} \pm 3^{\circ}$F. The driver shall be conditioned at the required temperature for a minimum of 6 hours immediately prior to testing. Failure of one or more drivers to meet requirements of 3.4.4 shall cause rejection of the lot.

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4.6.5 Jolt test (C3). The explosive drivers shall be mounted in the Jolt Test Fixture, LD 480002, and shall be subjected to Test 101 of MIL-STD-331. The occurrence of one or more drivers which explode or become unsafe to handle or dispose of as a result of the jolt test shall cause rejection of the lot.

4.6.6 Forty-foot drop test (C4). The explosive drivers shall be mounted in the test fixture shown on LD 480047. The test fixture containing the drivers shall be fastened to the carrier of the drop test apparatus. The drop test apparatus which is comprised of three basic components as follows shall be used for this test.

a. A rigidly-supported, hardened steel anvil having a minimum diameter of 6.0 inches and a minimum thickness of 2.0 inches.

b. A carrier to which the test fixture containing the drivers shall be fastened. The striking surface of the carrier shall have a spherical radius of 6.000 ± .050 inches, and shall have a minimum cross-sectional diameter of 2.5 inches at its greatest diameter, measured normal to the axis of the carrier. The axis of the carrier and the axis of the striking surface shall coincide within ± .050 inch.

c. A vertical guide which shall serve to guide the carrier in its drop to impact the anvil. The design of the guide surfaces shall be such that a negligible amount of friction between the carrier and guide is developed in order to simulate free fall as nearly as possible. The occurrence of one or more drivers which explode or become unsafe to handle or dispose of as a result of the 40-foot drop test shall cause rejection of the lot.

4.7 Acceptance criteria. Any Explosive Driver Mark 3 Mod 0 which fails to meet any of the performance requirements, product characteristics or environmental requirements of this specification shall be regarded as defective.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging (C5).

5.1.1 Level A.

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5.1.1.1 Cleaning, drying and preservative application. Not applicable.

5.1.1.2 Unit packaging. Unit packaging of the drivers shall be in accordance with Figures 1 and 2, steps 1, 2 and 3. The unit shall contain 25 drivers.

5.1.1.3 Intermediate package. Not applicable.

5.1.2 Levels B and C. Not applicable.

5.2 Packing.

5.2.1 Level A. In addition to the following, Level A packing shall be in accordance with the Code of Federal Regulations 49CFR 171-178.

5.2.1.1 Exterior containers. 500 drivers packaged as described in para 5.1.1.2 shall be packed in accordance with Figure 2, Step 4.

5.2.1.2 Levels B and C. Not applicable.

5.3 Marking.

5.3.1 Special markings. Markings of external containers shall be in accordance with Figure 2, step 4.

5.3.2 Normal markings. In addition to the markings required by contract or order, unit packages and shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

6. NOTES

6.1 Intended use. The Explosive Driver, Mark 3 Mod 0 is an electric ignition explosive dimple motor, used to provide mechanical actuation.

6.2 Ordering data. Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Government activity or contractor, as appropriate, to perform first article inspection and periodic production tests.

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c. When first article samples are waived because of having been approved on a recent contract or order, the purchase document shall include a statement that standards of workmanship exhibited by the previously submitted first article sample shall determine the minimum requirements of the contract or order relative to workmanship.

d. If the procurement document establishes periodic test sample sizes at variance with those required by this document, the procurement document shall also furnish a new test chart or in some manner designate the desired level and sequence of testing for each of the sample units.

e. That the safety precaution requirements of the Contractor's Safety Manual for Ammunition, Explosives and Related Dangerous Material, DOD 4145.26M, are applicable. NOTE: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of Ammunition Ashore, OP 5, should be made applicable.

6.3 Definitions. For the purpose of this specification, the following definitions apply:

6.3.1 Explosive driver. Reference to the driver, unit or explosive driver means Explosive Driver Mark 3 Mod 0.

6.3.2 Lot. Unless otherwise specified in the contract or purchase order, a lot size shall be one month's scheduled production. If a month's scheduled production is not an acceptable lot size, lot size then shall be specified.

6.4 Advisory handling of charges.

6.4.1 The loading room shall have a relative humidity of 70-80 percent. This moisture content shall be maintained at all times.

6.4.2 Only approximately 2 grams of charge shall be present at each loading station and only when an operator is present.

6.4.3 Any mixture which is spilled shall be cleaned up immediately by means of a cloth wet with butyl acetate; the soiled cloth shall be placed in a special disposal can.

6.4.4 The mixing and loading of the charge shall be done behind an approved shield whenever possible.

6.4.5 The clear lacquer and butyl acetate thinner that are mixed with lead styphnate confers a degree of resistance to explosion or ignition by impact or friction; however, it is emphasized that the resistance to explosion obtained by wetting in the prescribed manner is of a lower order than that obtained by water-wetting. Butyl acetate is flammable and can produce ignitible vapors.

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6.4.6 Constantly check the loading dish to prevent the mixture from becoming too dry, particularly the mixture around the edges and on the top of the dish. If necessary, the operator shall add Butyl Acetate, TT-B-838, and work the mixture to the proper loading consistency.

6.4.7 The clear lacquer and butyl acetate shall be mixed using a vulcanized fiber rod.

6.4.8 All utensils and tools shall be kept clean and free of any dried charge. Empty dishes shall be placed in approved containers.

6.4.9 Extreme care and precautions shall be exercised at all times in the handling and loading of the charge.

6.5 Advisory material. The following materials are believed to meet the requirements of this specification and are furnished as advisory information:

a. Lead, Grade MD-201	Metal Disintegrating Company, Elizabeth, New Jersey
b. Selenium, Rhedium Grade	R. T. Vanderbilt Company New York, New York
c. Nitrostarch	Trojan Powder Company, Allentown, Pa.

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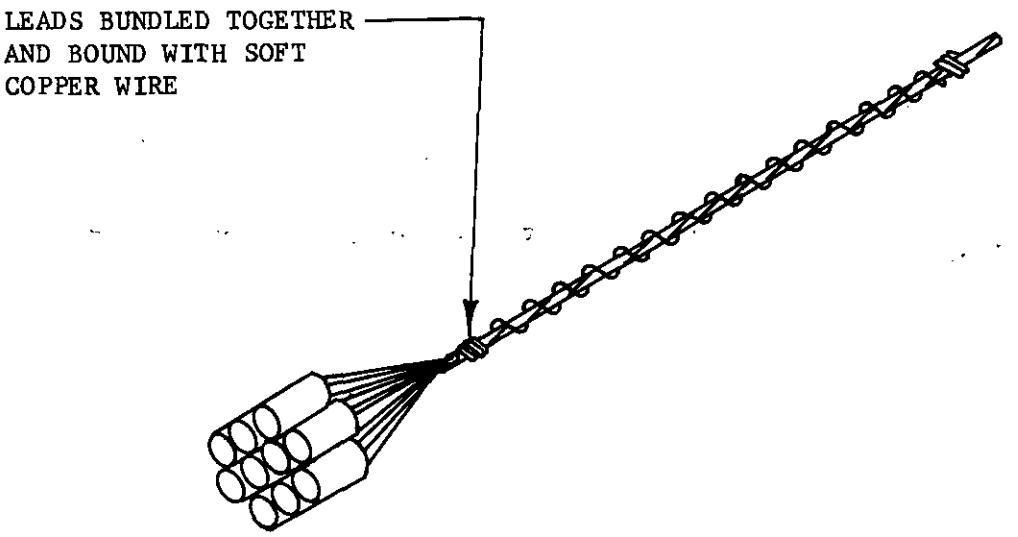
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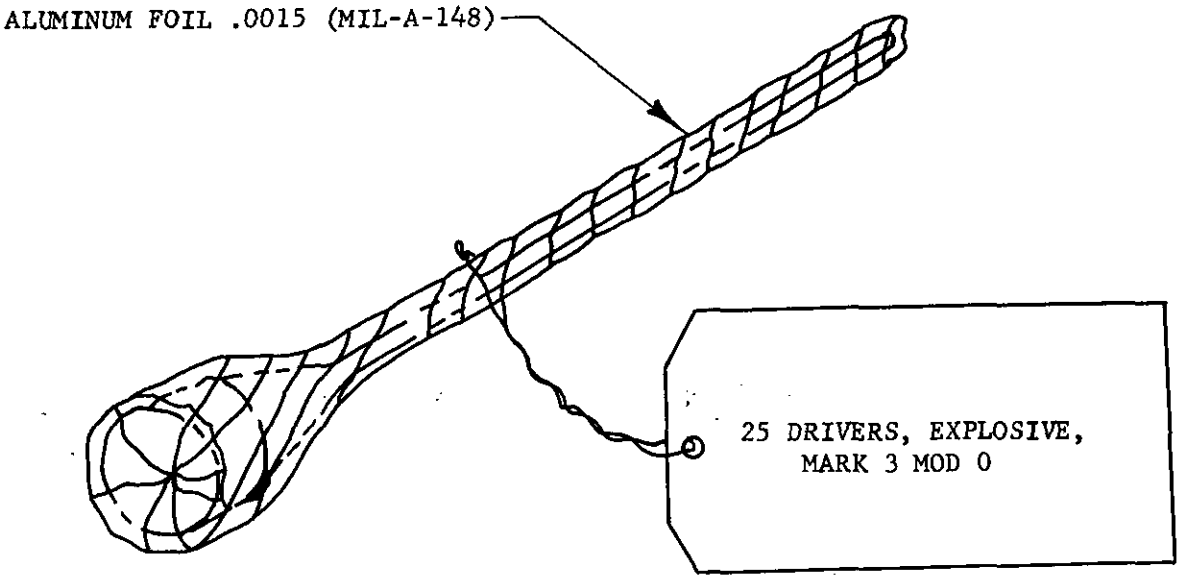
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FIGURE 1



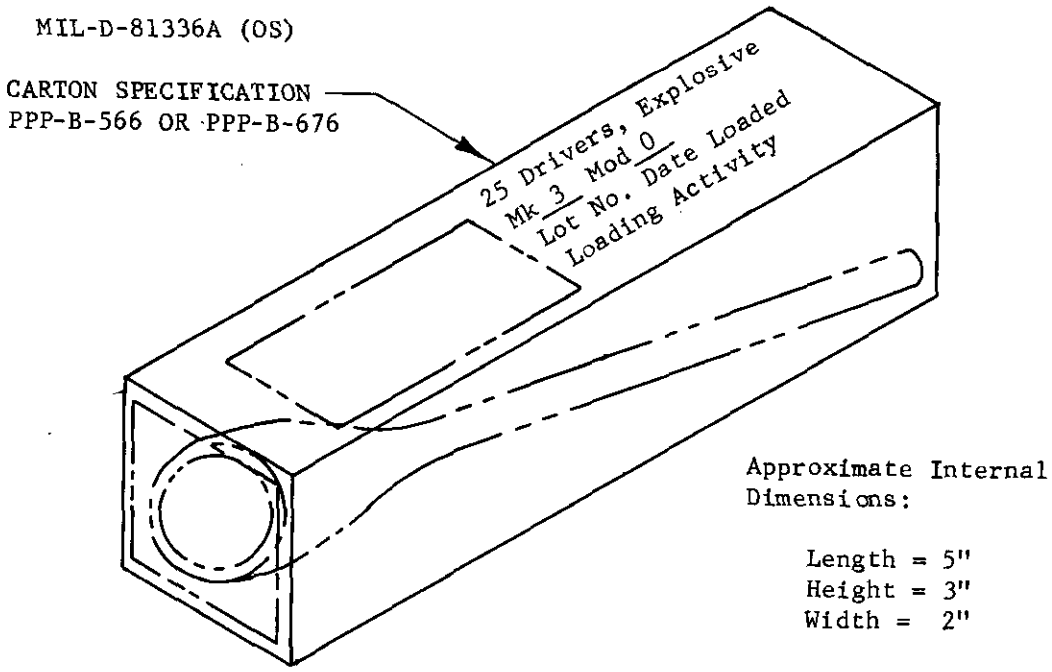
STEP 1



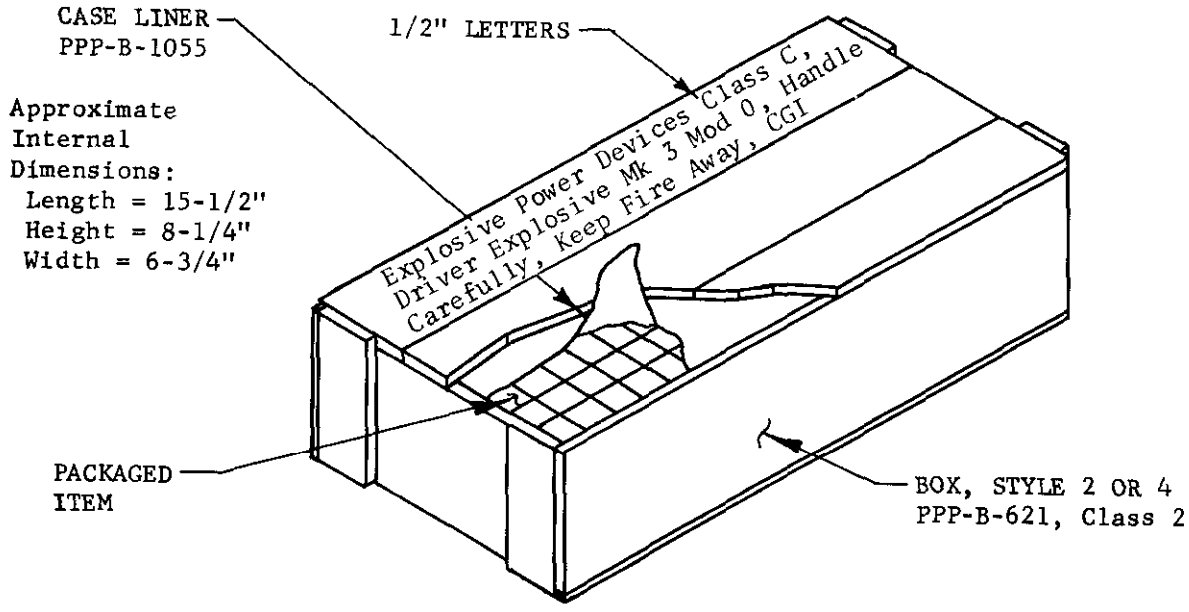
STEP 2

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FIGURE 2



STEP 3



STEP 4

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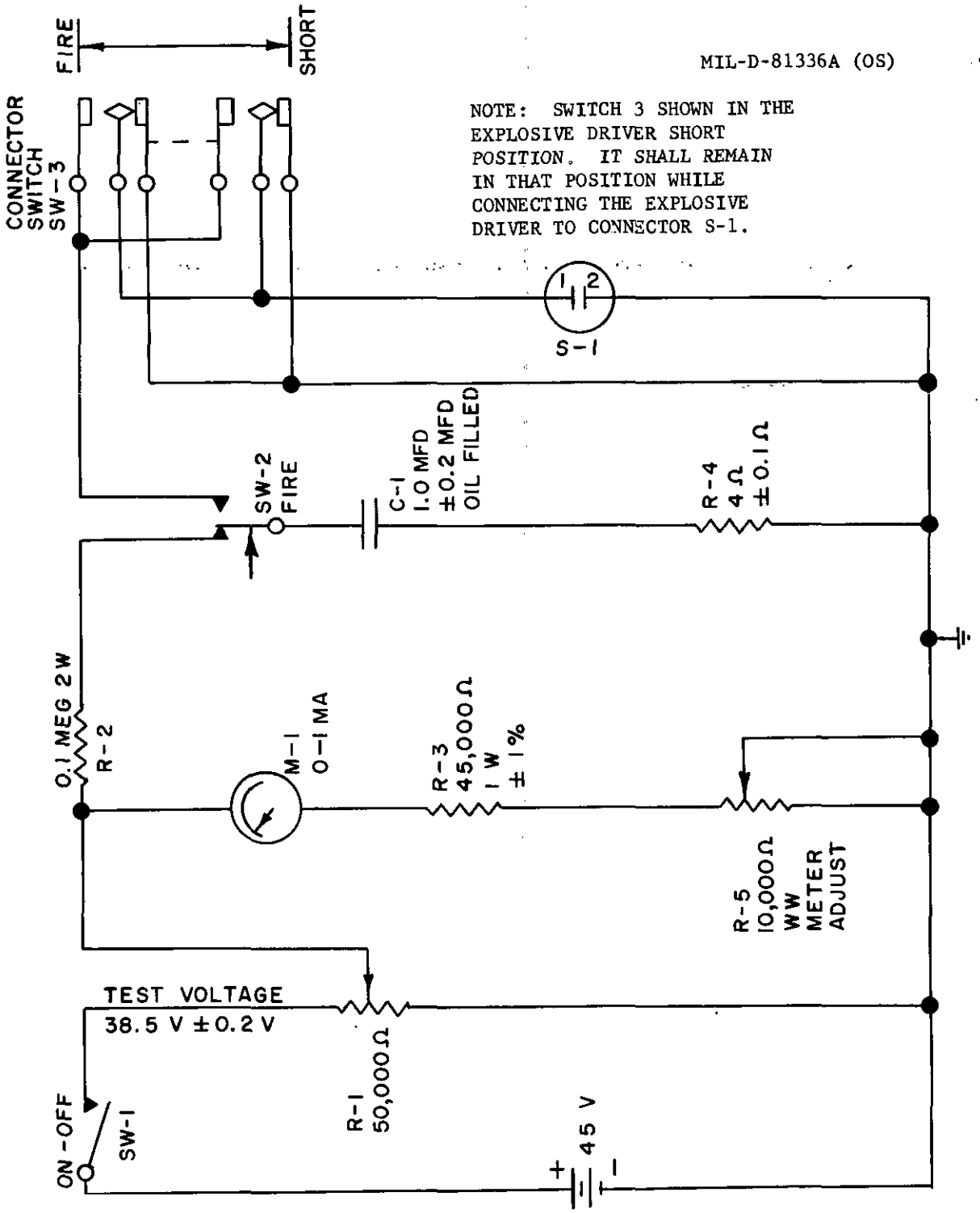


FIGURE 3

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SPECIFICATION ANALYSIS SHEET

Form Approved
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INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
MIL-D-81336A (OS), Driver, Explosive, Mark 3 Mod 0

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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